Introduction to Neuro-Symbolic AI

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt

March 16, 2026

My background

10/11 - 05/18	**University of Würzburg, TU Berlin** Basic training in mathematics and physics
06/18 - 12/18	**Fritz Haber Institute, Max Planck Society** Data-driven prediction of crystal properties
01/19 - 12/21	**Institute of Mathematics, TU Berlin** PhD in statistical learning theory
06/21 - 06/22	**InstruNEXT GmbH** Data-driven optimization of photomask baking processes
07/22 - 12/22	**Schneider Electric SE** AI-dedicated hardware
01/23 - 06/25	**DATEV eG** Automation of Accounting
since 06/25	**Weierstraß Institute for Applied Analysis and Stochastics** Neuro-symbolic AI, quantum machine learning

Outline

- Neuro-symbolic AI
- Use Case: DATEV Automation of Accounting
- Tensors in Artificial Intelligence
- Topics of the course

The future of the software industry

Traditional "symbolic" approach

- Software designed by hard-coded logic
- Requires structured input
- Human-interpretable behavior

AI-based "neural" approach

- Software consists in AI agents based on LLMs
- Flexible input handling
- Black box models

Aim of neuro-symbolic AI

Integrate both approaches:

- Human-interpretable model structure
- Flexibility by controlled LLMs

Strenghts and weaknesses of LLMs

Strengths of LLMs

- **Translation:** Translations between natural (e.g. english) and formal (e.g. python) languages
- **Generation:** Creation of new texts similar to existing corpus

Two major concerns about LLMs

Besides having astonishing memory capabilities, LLMs lack in

- **Explainability:** Problems with hallucinations and formal verifiability
- **Efficiency:** Massive power consumptions during reasoning

How to extend the capabilities of LLMs?

- State-of-the-art AI is built around the inference of LLMs
- But: Inference of LLMs is not the only thing, in which computers are good at!

Examples:

- Execution of an algorithm: Programming languages, e.g. python
- Retrieval of knowledge from a structured database: Query languages, e.g. SQL, SPARQL

Aim: Extend the capabilities of LLMs

Integrate inference of LLMs with inference in task-specific formal languages.

The neuro-symbolic AI initiative

Neural AI: Bringing efficiency

- Computations (gradients/inference) organized in layers (sets of neurons with tensor weights)
- Deep layers providing effective representation of data (task-dedicated neurons)

Symbolic AI: Bringing ante-hoc explainability

- Based on human interpretable logic
- Traditional approach to AI

How to bring both paradigms together?

Design neural architectures representing symbolic reasoning.

Outline

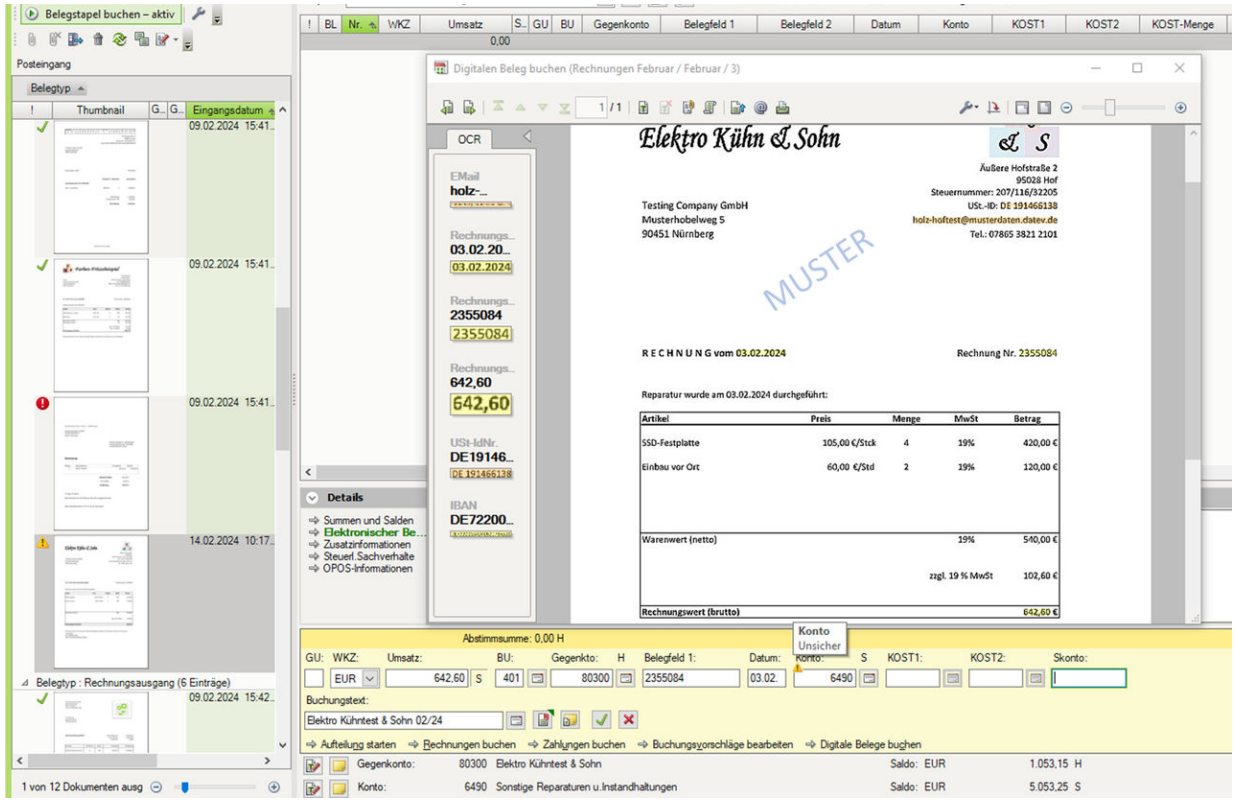
- Neuro-symbolic AI
- Use Case: DATEV Automation of Accounting
- Tensors in Artificial Intelligence
- Topics of the course

DATEV Automation of Accounting

DATEV, a software company from Nuremberg, sells automation solutions for accounting consultants

- Consultant receives invoices from its customers
- Accounting statements need to be generated

Screenshot from the product[1]:



[1] https://www.datev.de/web/de/steuerberatung/loesungen/rechnungswesen/buchfuehrung-erstellen/automatisierungsservices-rechnungswesen/datev-automatisierungsservice-rechnungen

Requirements

Performance by Neural AI:

- Handle most cases without the need of human supervision
- Understand scenarios based on world knowledge (e.g. "Lindt" in a position indicates choclate consumption)

Trust and manipulations by Symbolic AI:

- Customers need to guarantee certain behaviour, such as respectation of laws
- Manipulation of models based on human input

This requires a form of neuro-symbolic AI!

Toy Accounting Example

To showcast the approach we study a toy accounting example. For each invoice three variables are either True or False:

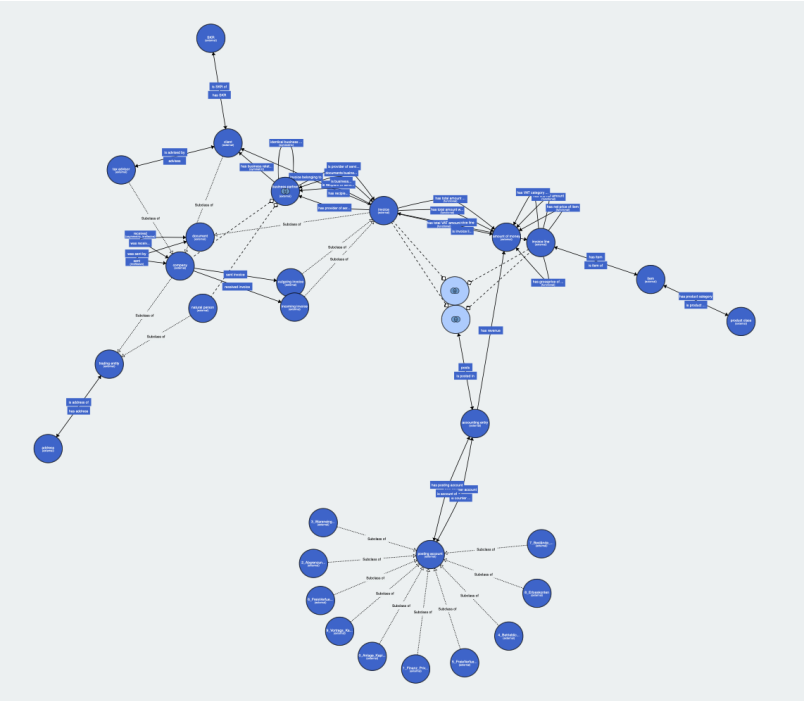
- A Feature is present on the receipt (e.g. whether the receipt was received or sent by the customer).
- The account Account 1 is used to book the receipt.
- The account Account 2 is used to book the receipt.

Let there be two accounting rules to be respected:

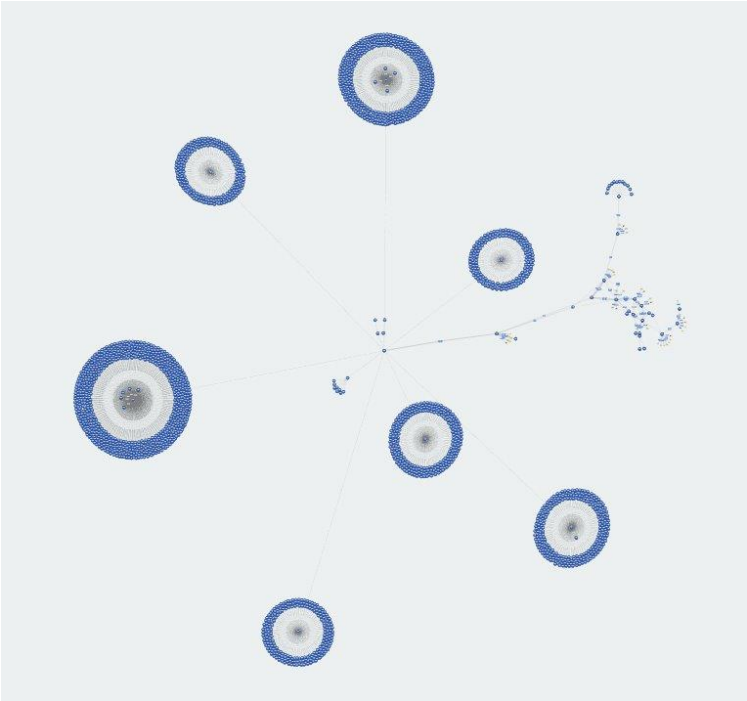
- Exactly one of the accounts Account 1 and Account 2 is used to book the receipt.
- When the Feature is present on the receipt, the Account 1 needs to be booked.

Realistic Problem Sizes

The variables and main relations are visualized in a so-called ontology[2]:



Coarse Visualization



Detailed Visualization

We will come back to ontologies and knowledge graphs later in the course.

[2] https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions

Outline

- Neuro-symbolic AI
- Use Case: DATEV Automation of Accounting
- Tensors in Artificial Intelligence
- Topics of the course

Representation of the toy accounting system

The system has a representation by the variables taking values in $\{0, 1\}$ (i.e. $m = 2$):

- X_F: Whether the Feature is present on the receipt.
- X_{A1}: Whether the Account 1 is used.
- X_{A2}: Whether the Account 2 is used.

The system states are the cartesian product of the variable states $\{0, 1\}$:

$$\bigtimes_{k \in \{F, A1, A2\}} \{0, 1\} = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

The toy accounting rules exclude some states:

- Exactly one of the accounts Account 1 and Account 2 is used to book the receipt: $X_F \Rightarrow X_{A1}$
- When the Feature is present on the receipt, the Account 1 needs to be booked: $X_{A1} \oplus X_{A2}$

Coordinatewise tensor visualization

Boolean tensors of leg-dimension $m = 2$ are

$$\tau \; : \bigtimes_{k \in \{0,\ldots,d-1\}} \{0,1\} \to \mathbb{R}\,.$$

We visualize such maps as

- $d = 1$: **Vectors**

$$X_{A1} : \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- $d = 2$: **Matrices**

$$X_{A1} \oplus X_{A2} : \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad , \quad X_F \Rightarrow X_{A1} : \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

- $d = 3$: **Order-3 tensors**

$$(X_{A1} \oplus X_{A2}) \wedge (X_F \Rightarrow X_{A1}): \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Tensors: Natural description of factored representations

Factored representations are collection of variables $\{X_0, \ldots, X_{d-1}\}$, each with a finite set of states $\{0, \ldots, m_k - 1\}$ (so far: $\{0, 1\}$ for $m = 2$). The set of states is then the cartesian product

$$\bigtimes_{k \in \{0,\ldots,d-1\}} \{0, \ldots, m_k - 1\}$$

Knowledge about the system, for example the possibility of each state, is a tensor

$$\tau : \bigtimes_{k \in \{0,\ldots,d-1\}} \{0, \ldots, m_k - 1\} \to \mathbb{R}$$

Tensors for neuro-symbolic AI

Tensors (and their networks) are a unifying language for Artificial Intelligence

- **Logical knowledge bases:** Boolean coordinates, decomposition by logical syntax
- **Probability distributions:** Non-negative coordinates, decomposition by independencies
- **Neural networks:** Representation of arbitrary function decompositions
- **Data bases:** Relations stored in tables are tensors

Tensors capture the

- Structure of symbolic AI
- Flexibility of neural AI

We exploit the tensor formalism in this course for neuro-symbolic AI.

Tensors for neuro-symbolic AI

- **Modelling principle:** Representation of logical and probabilistic models by tensors.
- **Sparsity principle:** Decomposition of tensors into tensor networks.
- **Inference principle:** Contractions of tensor networks.

The course orients on the preprint:

A tensor network formalism for neuro-symbolic AI

Alex Goessmann,* Janina Schütte, Maximilian Fröhlich, and Martin Eigel

Weierstrass Institute of Applied Analysis and Stochastics
Anton-Wilhelm-Amo-Straße 39
Berlin, 10117, Germany

*alex.goessmann@wias-berlin.de

[cs.AI] 21 Jan 2026

Abstract

The unification of neural and symbolic approaches to artificial intelligence remains a central open challenge. In this work, we introduce a tensor network formalism, which captures sparsity principles originating in the different approaches in tensor decompositions. In particular, we describe a basis encoding scheme for functions and model neural decompositions as tensor decompositions. The proposed formalism can be applied to represent logical formulas and probability distributions as structured tensor decompositions. This unified treatment identifies tensor network contractions as a fundamental inference class and formulates efficiently scaling reasoning algorithms, originating from probability theory and propositional logic, as contraction message passing schemes. The framework enables the definition and training of hybrid logical and probabilistic models, which we call Hybrid Logic Networks. The theoretical concepts are accompanied by the python library tnreason, which enables the implementation and practical use of the proposed architectures.

Tensors: A computational interface for Machine Learning

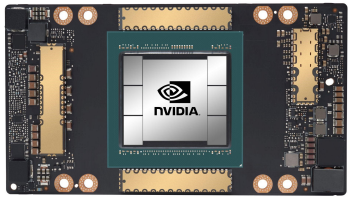
On the software side:

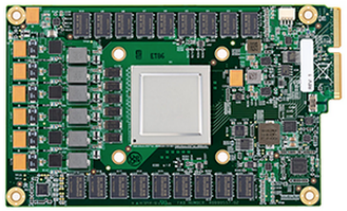
- Parallel computing enabled through (multi-)linear algebra
- Tensors are main classes is standard ML libraries: TensorFlow, Pytorch

On the hardware side:

- Dedicated for graphic processing: Graphic Processing Units (GPUs)
- AI-dedicated hardware: Tensor Processing Units (TPUs)

TensorFlow





Outline

- Neuro-symbolic AI
- Use Case: DATEV Automation of Accounting
- Tensors in Artificial Intelligence
- Topics of the course

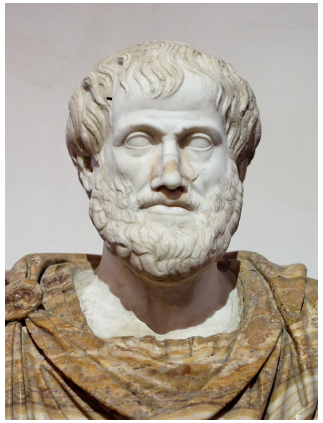
I - Logical Approaches

Logic as the theory of human reasoning:

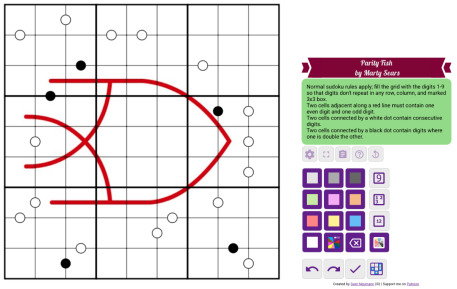
- Dating back to greek philosophers modelling Human Intelligence
- Traditional approaches to Artificial Intelligence

Logics consists in

- **Syntax:** Description of knowledge
- **Semantics:** Interpretation of knowledge, inference of new knowledge



Aristotle[a]



Modified Sudoku puzzles[b]

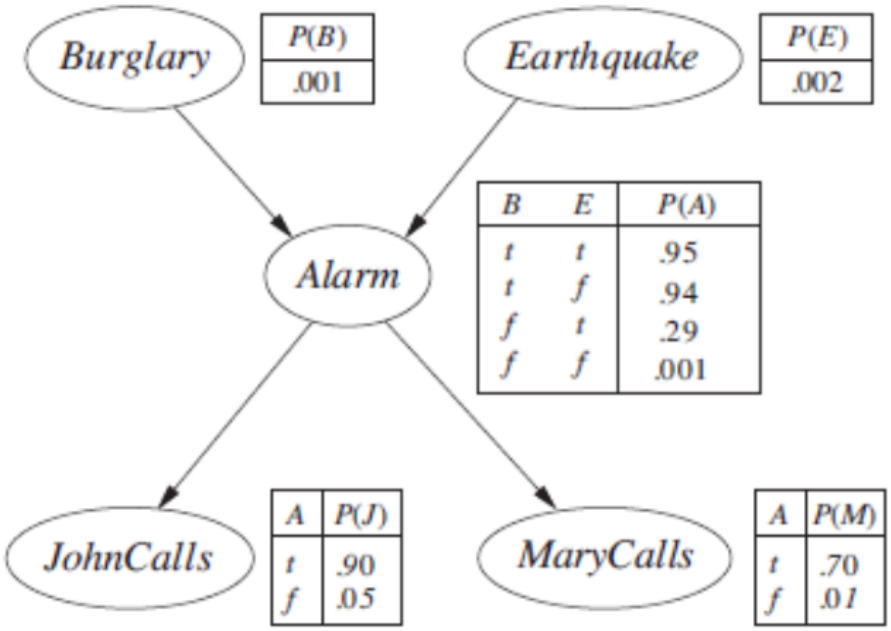
Challenges

- Find sparse representation mechanisms
- Efficient reasoning schemes based on tensor network contractions

[a] https://de.wikipedia.org/wiki/Aristoteles
[b] https://sakana.ai/sudoku-bench/

II - Probabilistic Approaches

Bayesian Networks model probabilistic relations between variables



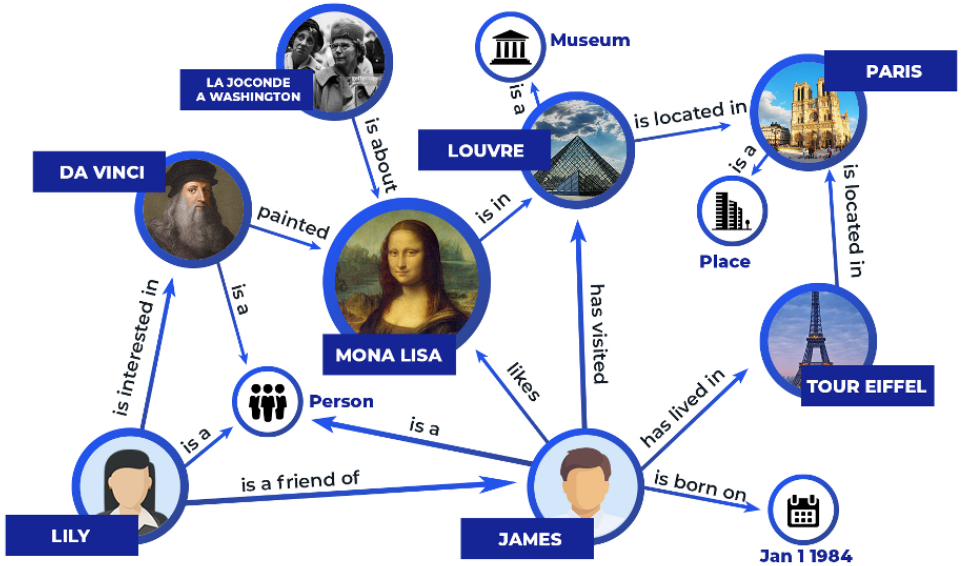
Challenges

- Learn the dependence structure of the variables
- Extract the causal relations of the variables

III - Statistical Models of Knowledge Graphs

Extension from factored representations to structured representations:

- Worlds contain objects from classes.
- The objects are enumerated using further tensor axes.



[3]

Challenges

- Reason about the relations in a knowledge graph
- Predict the connectivity of new objects

[3] https://ml-digest.com/knowledge-graphs/

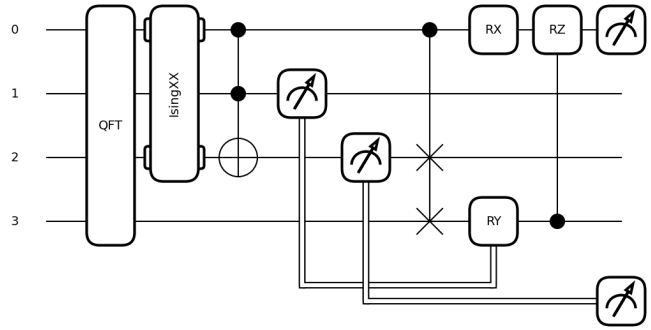
IV - Quantum Machine Learning

Quantum computation

- Formulated in tensor spaces
- Quantum circuits: Tensor networks

Quantum machine learning

- Inference in quantum circuits
- Promise: Efficiency increase (to be shown)



Example of a Quantum Circuit[a]:

[a] https://docs.pennylane.ai/en/stable/introduction/inspecting-circuits.html

Challenges

- Find sparse quantum circuits exploiting a models sparsity
- Increase the efficiency of reasoning with tensor networks by quantum computation.

Further literature

Introductory textbooks:

- **Russel, Norvig: Artificial Intelligence - A Modern Approach** (Fourth Edition), Pearson Education 2021
- **Koller, Friedman: Probabilistic Graphical Models: Principles and Techniques**, MIT Press 2009
- **Antoniou et al: A Semantic Web Primer** (Third Edition), MIT 2012
- **Schuld, Petruccione: Machine Learning with Quantum Computers**, Springer 2021

Advanced textbooks:

- **Wainwright, Jordan: Graphical Models, Exponential Families and Variational Inference**, now 2008
- **Hackbusch: Tensor Spaces and Numerical Tensor Calculus**, Springer 2019